Date : ....

QUESTION 1 :

Write a program to make a suitable class to input two integers. Interchange the integers. Print integers before and after interchange with suitable message.

Answer :

```
class A
{
    int num1, num2;
    void main()
    {   num1=5; num2=9; int a=num1, b=num2;
        num1 = num1+num2;
        num2 = num1-num2
        num1 = num1-num2;

        SOP(" Before Interchange");
        SOP(" num1 =" + a);
        SOP("num2 =" + b);
        SOP("\nAfter Interchange");
        SOP(" num1=" + num1);
        SOP(" num2=" + num2);
    }
}
```

CARE TO BE TAKEN WHILE WRITING THE PROGRAMS TO AVOID ERRORS.

1. Documentation / remarks must be included with each program for better understanding.
2. The class, function(s) and variables with data types (String, *char, short, int, long, float, double and boolean*) must be as per their syntaxes.
3. You may use both **Scanner class** and **BufferedReader** input streams to input/read data from the keyboard.
3. You may run the program either using BlueJ or JDK. In BlueJ void main( ) function is not needed, you can run the program by selecting the required function(s). But in JDK main( ) function is must. Using import java.io.*; is optional but compulsory if the program contains input statements (refer sample question 1, without import statement and sample question 2 with import statement).
4. Variable description must be included with every program in three columns via-**variable-name**, **data-type** and **purpose** of the variable.

Date : ....

QUESTION 2 :

Write a program to input name of candidate (string), roll number (long integer), date of examination (string) and examination center (string). Print the admit card of the candidate in the following format :

THE OFFICERS SELECTION COMMISSION

ADMIT CARD

NAME OF CANDIDATE : DATE :

EXAMINATION CENTER : ROLL NUMBER :

Answer :

```
import java.util.*;
class A
{
  String name, DOE, EC;
  long RN;
  void main()
  {
    Scanner sc=new Scanner(System.in);
    SOP("Enter name, Roll No, DOE, EC");
    name = sc.nextLine()
    DOE  = sc.nextLine();
    EC   = sc.nextLine();
    RN   = sc.nextLong();
    SOP("\t\t\t\t The officers Selection Commision \n");
    SOP ("\t\t\t\t\t\t Admit Card \n");
    SOPt("Name of candidate : " + name);
    SOPt("\tDate : " + DOE);
    SOPtn("Examination Center: " + EC);
    SOPt ("Roll no:" + RN);
  }
}
```

Date : ....

QUESTION 3 :

Define a class **employee** with its data member **basic**. Find the gross pay of an employee for the following allowances and deductions. Use meaningful variables. Print name, basic pay, net pay and gross pay.

Dearness Allowance = 25% of Basic Pay.
House Rent Allowance = 15% of Basic Pay.
Provident Fund = 8.33% of Basic Pay.
Net Pay = Basic Pay + Dearness Allowance + House Rent Allowance.
Gross Pay = Net Pay - Provident Fund.

Answer :

```
class employee
{
    double basic;
    void main(double basic1, String Name)
    {
        basic = basic1;
        double DA= 25/100.0 * basic;
        double HRA= 15/100.0 * basic;
        double PF = 8.33/100 * basic;
        double NP = basic + DA + HRA;
        double GP = NP- PF;

        SOP ("Name :" + Name);
        SOP ("Basic :" + basic);
        SOP ("Net Pay:" + NP);
        SOP ("Gross Pay:" + GP);
    }
}
```

Date : ....

QUESTION 4 :

Define a class **Salary** described as below :

Data Members: Name, Address, Phone, Subject Specialization, Monthly Salary, Income Tax.

Member methods :

(i) To accept the details of a teacher including the monthly salary.

(ii) To display the details of the teacher.

(iii) To compute the annual Income Tax as 5% of the annual salary above Rs. 1,75,000.

Write a main method to create object of a class and call the above member method.

ICSE 2007

Answer :

```
import java.util.*;
class Salary
{
    String Name, Address, ~~Ph~~ SS;
    long Phone;
    double Salary, IT;

    void getDetails()
    {
        Scanner SC = new Scanner(System.in);
        SOP("Enter name, Address, SS, phone, Salary");
        Name = sc.nextLine();
        Address = sc.nextLine();
        SS     = sc.nextLine();
        Phone  = sc.nextLong();
        Salary = sc.nextDouble();
    }
```

```
void print.Details()
{
    SOP("Name: " + Name + "\n Address: " + Address +
        "\n SS: " + SS + "\n Phone: " + Phone +
        "\n Salary: " + Salary + "\n IT: " + IT);
}

void calcIT()
{
    double AS = Salary *12;
    if (AS > 175000)
        IT = 5/100.0 * AS)
}

void main()
{
    Salary S1 = new Salary();
    S1.accept
    S1.calcIT();
    S1.printDetails();
}
}
```

Date : ....

QUESTION 5 :

Write a program to input roll number of a student, marks in three subjects. Calculate total marks and percentage. Print the output using all input data and calculated data with proper headings. Use proper documentation in the program.

Answer :

```
import java.util.*;
class A
{
    int RN, m1, m2, m3;
    double TM, per;

    void main()
    {
        Scanner SC = new Scanner(System.in);
        SOP("Enter Roll No, Marks1, marks2, marks3");
        RN = SC.nextInt();
        m1 = SC.nextInt();
        m2 = SC.nextInt();
        m3 = SC.nextInt();
        TM = m1 + m2 + m3
        per = TM /300 * 100

        SOP("Roll No: " + RN + "\nMark1: " + m1 +
            "\nMark2: " + m2 + "\nMark3: " + m3 +
            "\nTotal Marks: " + TM +
            "\nPercentage: " + per);
    }
}
```

QUESTION 6 :

Write a class **consumerLIC** with **name** (consumer name), **pno** (policy number), **pramt** (premium amount) and **salary** as its data members. Find the gross salary of consumer as per the given criteria. Print name, pno, pramt, salary and gross salary.
Premium amount = 15% of salary.
Gross salary = salary - Premium amount

Answer :

```
class consumerLIC
{   string name, policyNo;
    double pramt, salary;

    void main (String n, String pN, double p, double s)
    {
        name = n;
        policyNo = pN;
        pramt = p;
        salary = S;
        pramt = 15/100.0 * salary;
        double GS = salary - pramt;

        SOP("Name:" + n + "\n PNo:" + policyNo +
            "\nPremium:" + pramt +
            "\n salary:" + salary +
            "\n Gross Salary:" + GS);
    }
}
```

Date : ....

QUESTION 7 :

Write a program to perform the following operations using mathematical methods:

(i) Print 6 raised to the power 3.

(ii) Print the result of 16.32 using ceil( ).

(iii) Print the result of 29.42 using floor( ).

(iv) Print the rounded off value of 136.662.

(v) Print the cube root of 256.

Answer :

```
class A
{
    void main()
    {
        SOP ( Math.pow(6, 3));
        SOP ( Math.ceil(16.32));
        SOP ( Math.floor(29.42));
        SOP ( Math.round (136.662));
        SOP ( Math.cbrt (256));
    }
}
```

Date : ....

QUESTION 8 :

Write a program to input three integers and print the largest and smallest integers using suitable mathematical functions.

Answer :

```
class A
{
    void main(int n1, int n2, int n3)
    {
        SOP("Largest :" + Math.max(n1, Math.max(n2,n3)));
        SOP("Smallest :" + Math.min(n1, Math.min(n2,n3)));
    }
}
```

QUESTION 9 : Date : ....

Write a program to input integers **X**, **Y** and **Z**. Using suitable mathematical methods print **X** raised to the power **Y**, square root and cube root of **Z**.

Answer :

```
class A
{
    void main(int x, int y, int z)
    {
        SOP("x power y:" + Math.pow(x,y));
        SOP("Sqrt of z:" + Math.sqrt(z));
        SOP("Sqrt of y:" + Math.sqrt(y));
        SOP("Cbrt of z:" + Math.cbrt(z));
        SOP("Cbrt of y:" + Math.cbrt(y));
    }
}
```

Date : ....

QUESTION 10 :

Write a program to input marks in three subjects in **m1**, **m2** and **m3**. Print the highest and lowest marks.

Answer :

```
class A
{
    void main(int m1, int m2, int m3)
    {
        SOP("Highest Marks:" + Math.max(m1, Math.max
                                (m2, m3)));

        SOP("Lowest Marks:" + Math.min(m1, Math.
                            min(m2, m3));
    }
}
```